



ChatGPT

- ❑ Chat Generative Pre-trained Transformer
- ❑ ChatGPT was launched on November 30, 2022, by San Francisco-based OpenAI.
- ❑ ChatGPT gained 1 million users in under a week

Introduction

APPLE

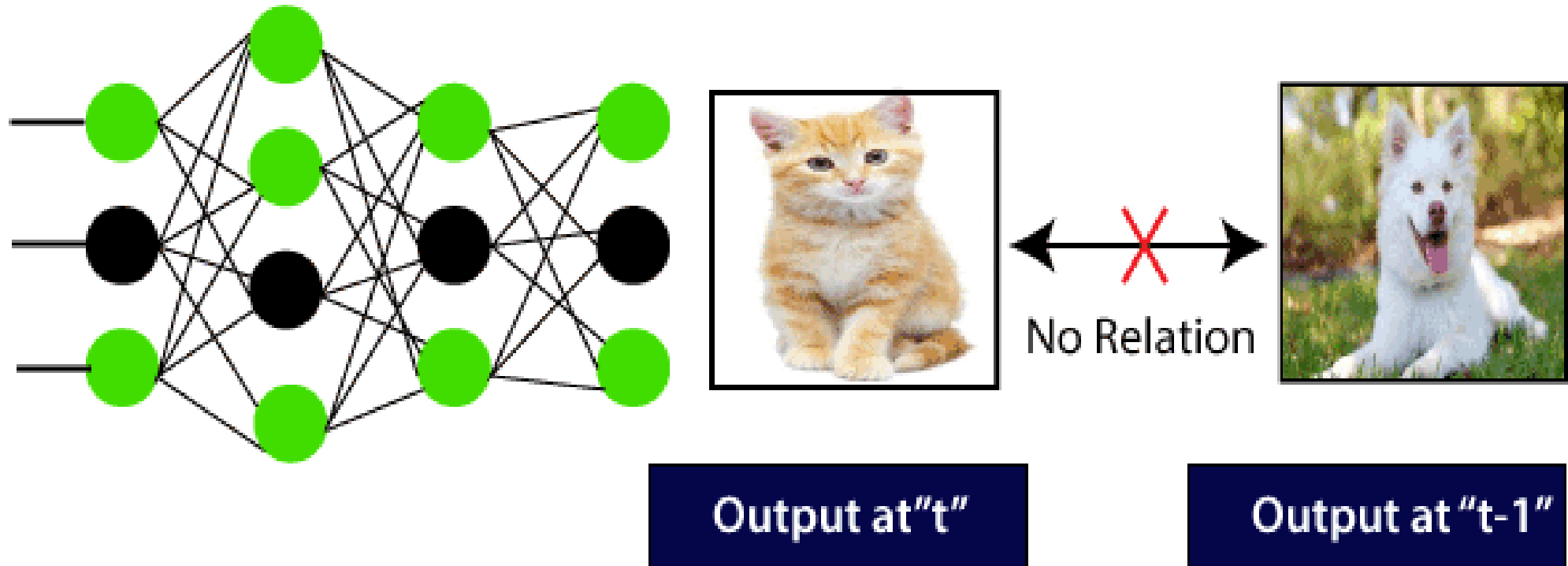


BALL



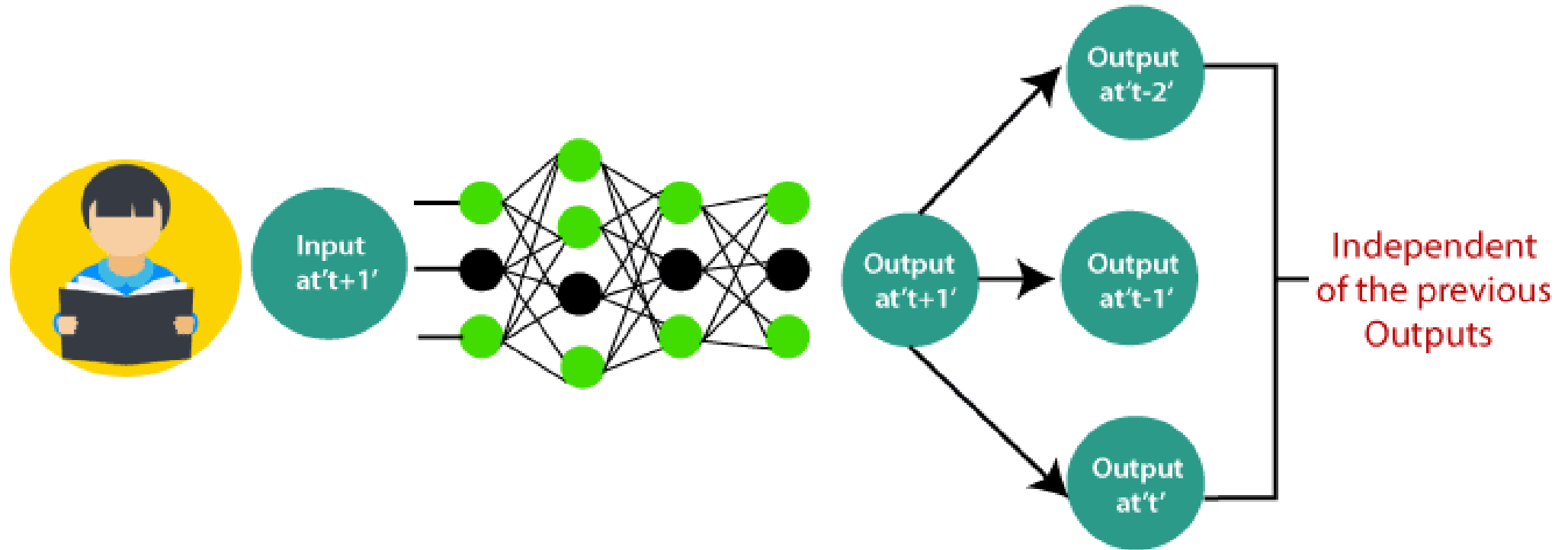
different features

Introduction



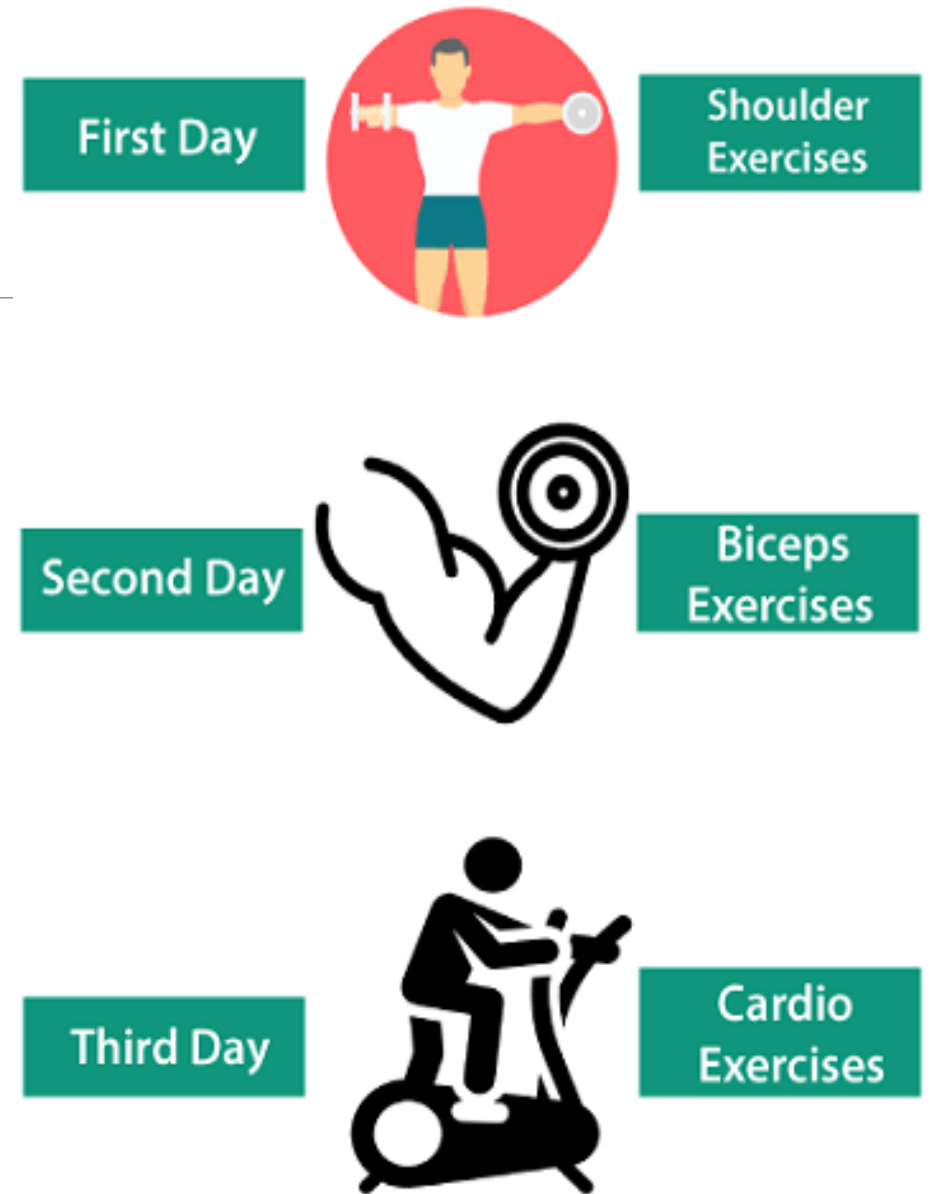
independent features

Introduction



Sequential Data Example

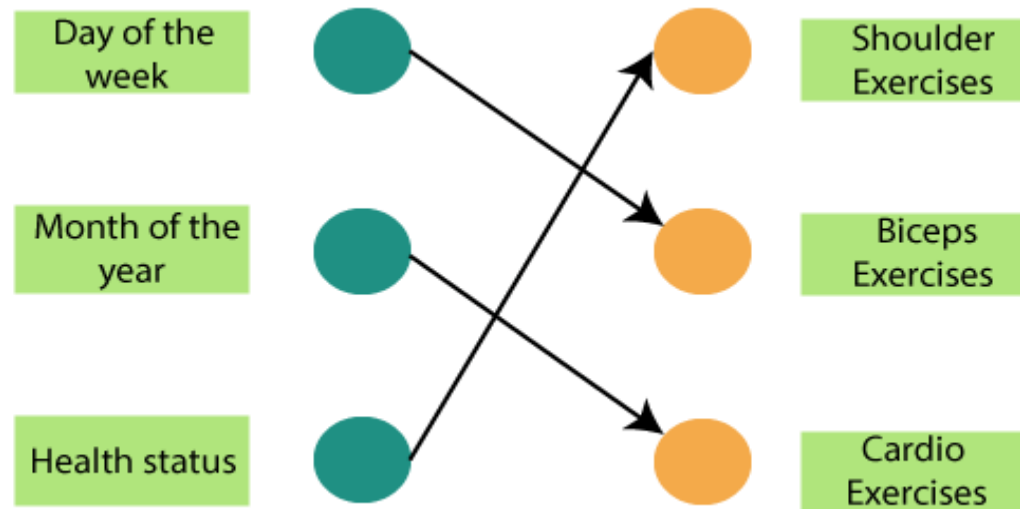
We go to the gym regularly,
and
the trainer has given us the schedule of our workout



Sequential Data Example

Note that all the exercises are repeated in a proper order every week.

Let us use a feed-forward network to trying and predicting the types of exercises.

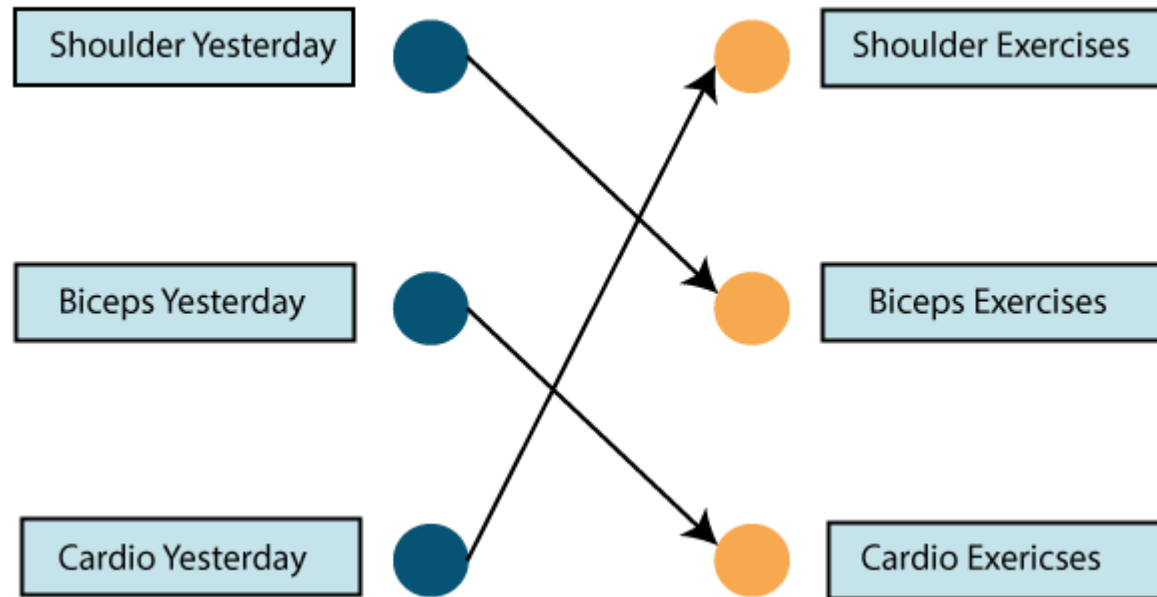


The inputs are the **day**, **month**, and **health status**.

Sequential Data Example

However, this will not very accurate, considering the input.

In this case, find the inputs to be workout done on the previous day.

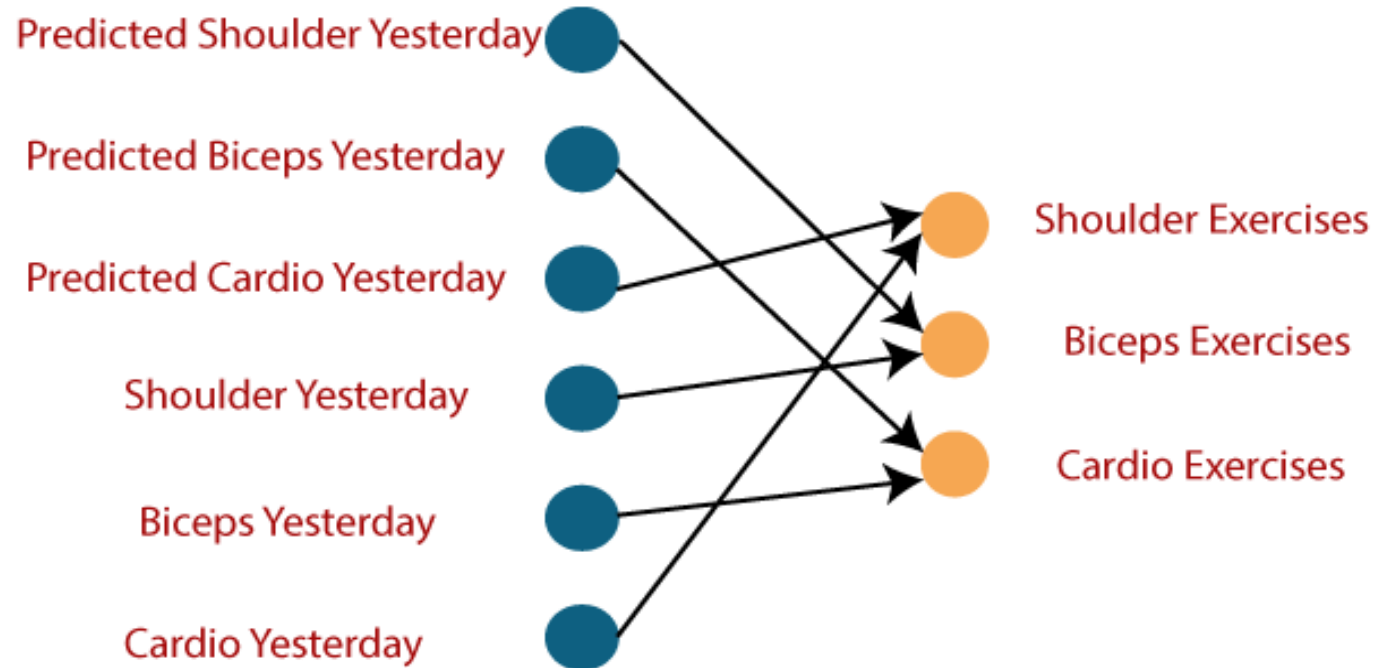


So if we did a **shoulder exercise** yesterday, we could do a **bicep exercise** today, and this goes on for the rest of the week.

But What if we happen to miss a day at the gym

Sequential Data Example

- If we happen to miss a day at the gym, the data from the previously attended timestamp can be considered.

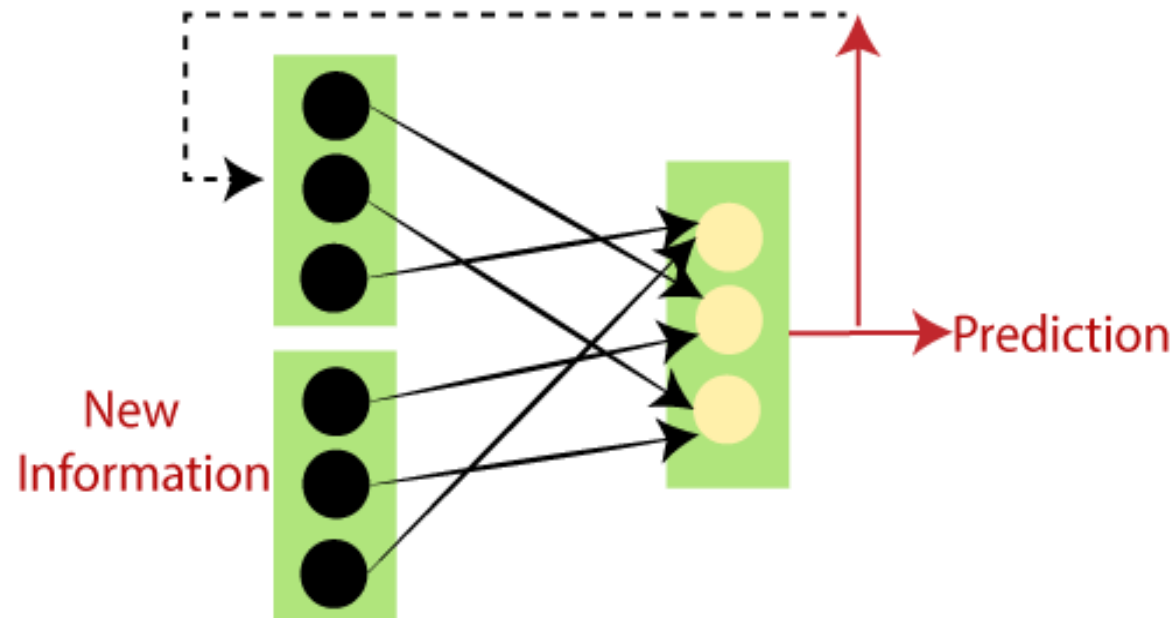


Sequential Data Example

- ❑ If a model is experienced based on the data it obtains from the last exercises, the output from the model will be accurate.
- ❑ To sum it, let us convert the data into vectors.
- ❑ Where vectors are numbers which are input to the **model** to **denote** if we have done the exercise or not.

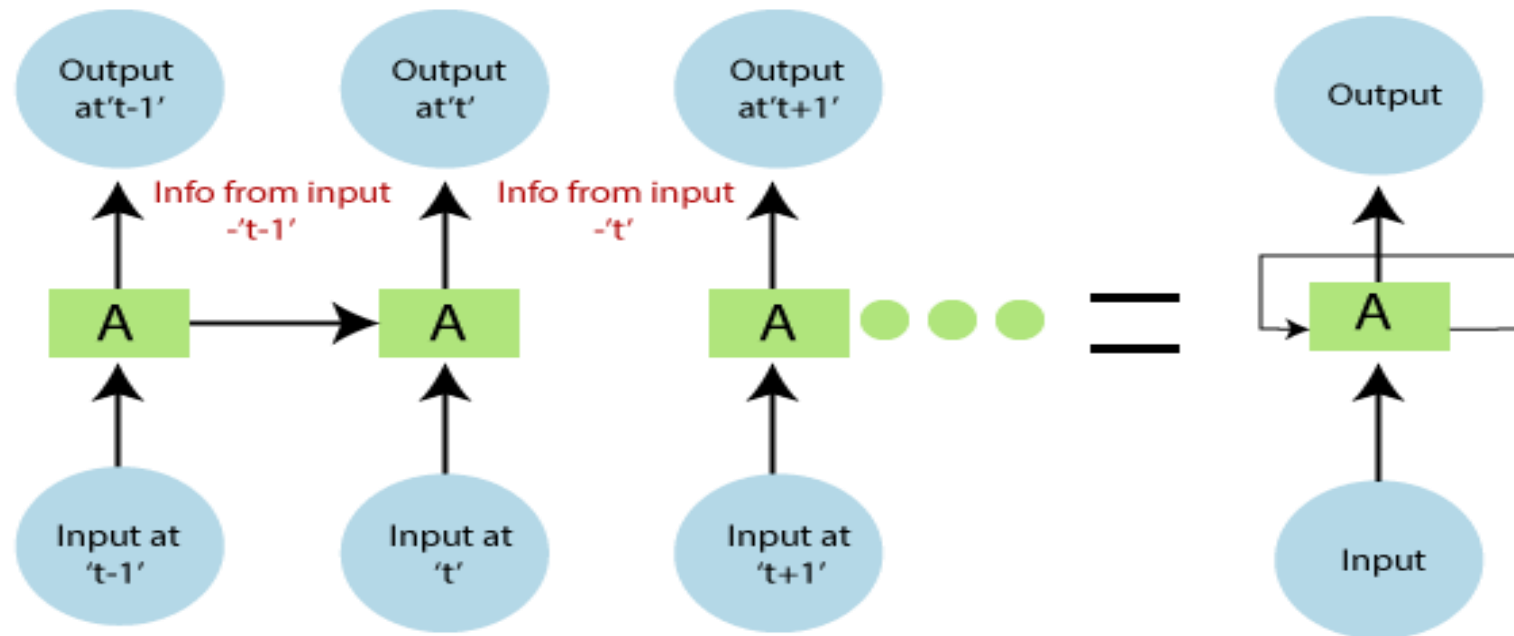
Sequential Data Example

- So, if we have a shoulder exercise, the corresponding node will be '1', and the rest of the exercise nodes will be mapped to '0'.



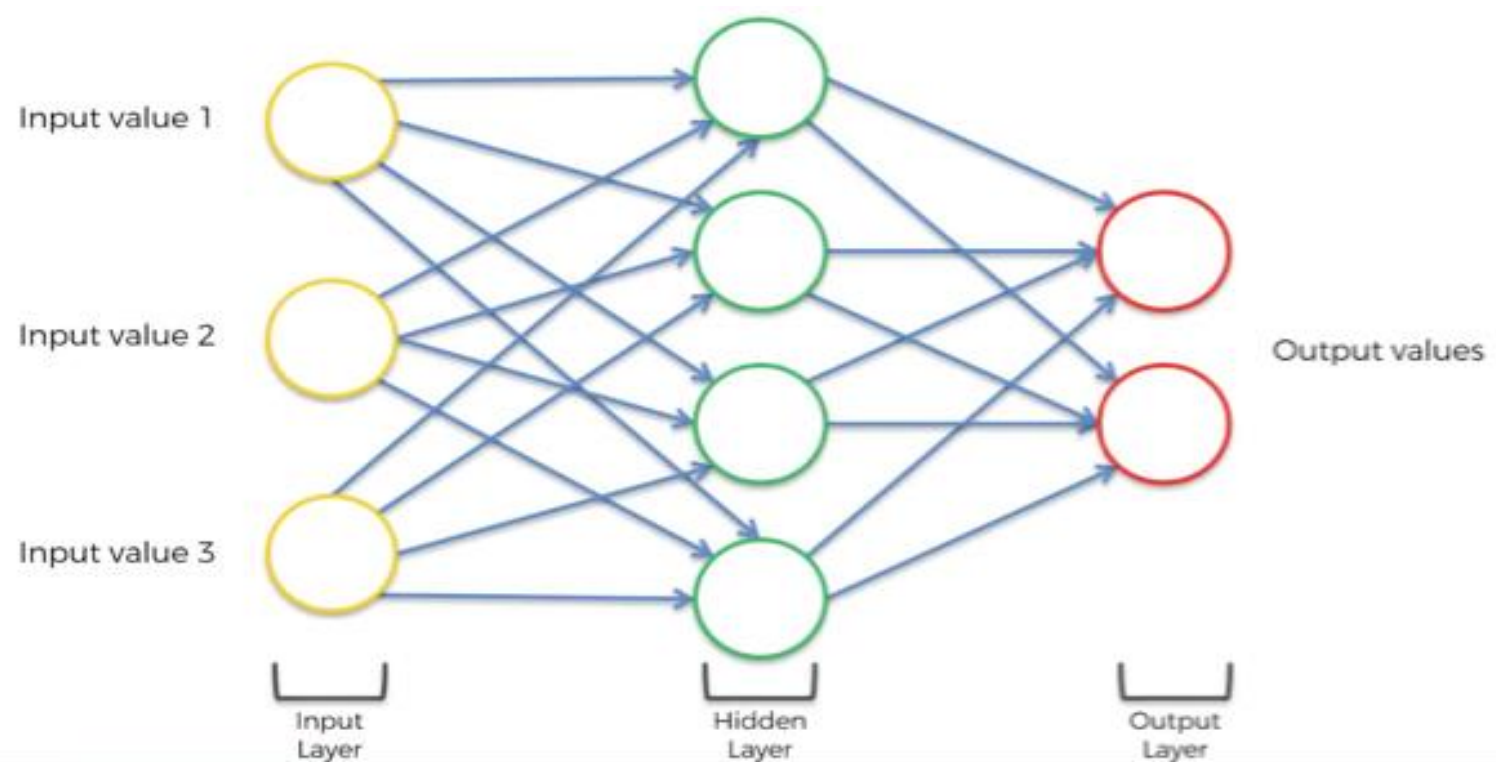
Recurrent Neural Network - RNN

- The recurrent network is a type of artificial neural network which is designed to recognize patterns in sequences of data. Like, text, genomes, handwriting, the spoken word, numerical times series data from sensors, stock markets, and government agencies.



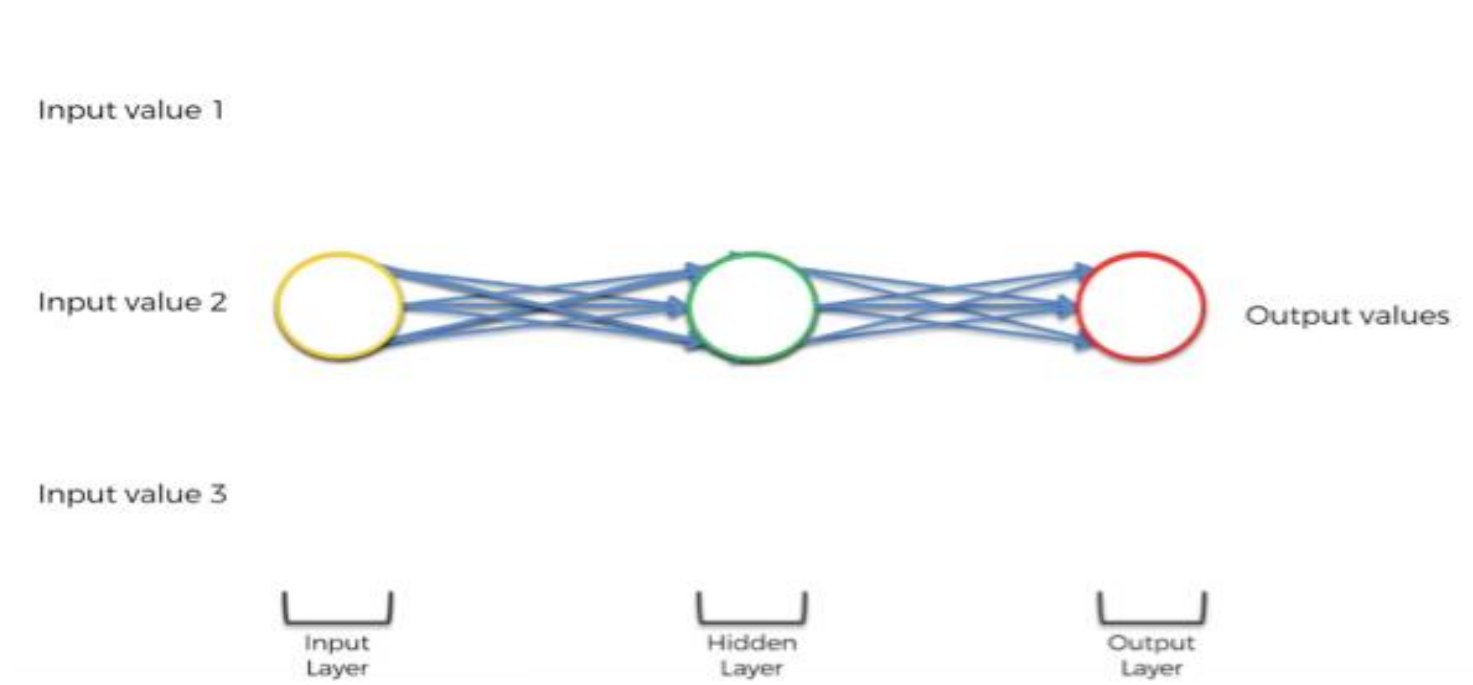
Representation of the RNNs

- We begin with a transformation of a simple ANN showed below into RNN.



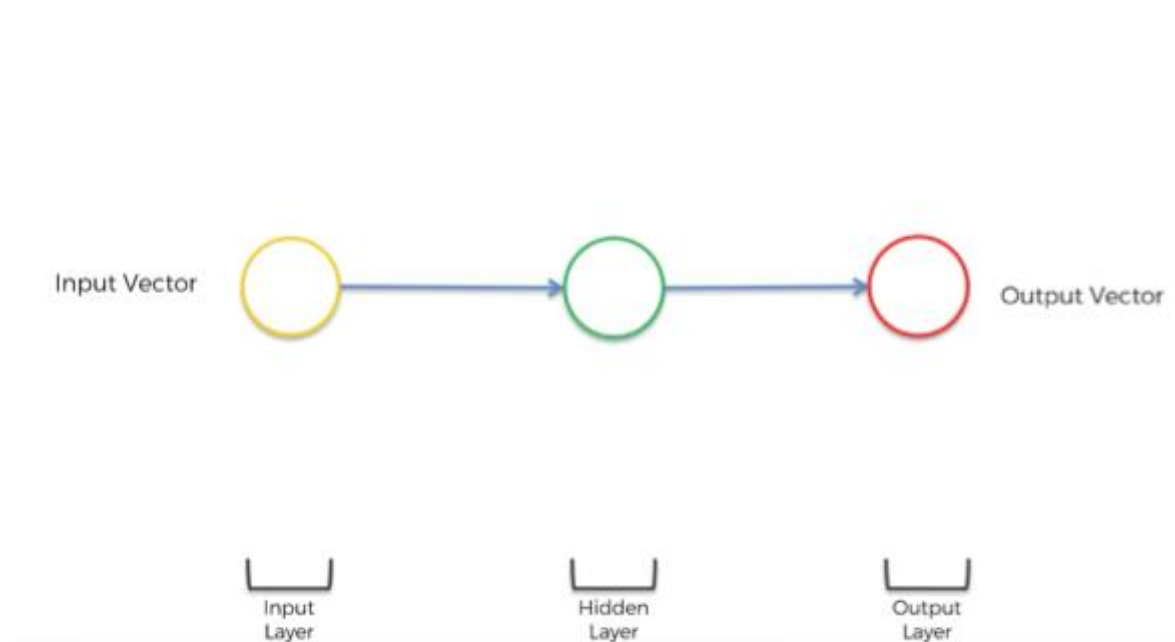
Representation of the RNNs

- ❑ Squashing the network. The layers are still there but think of it as if we're looking from underneath this neural network.



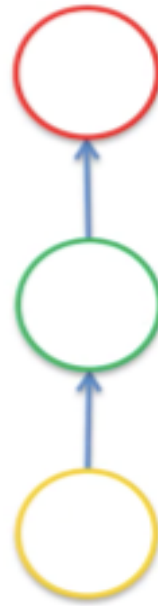
Representation of the RNNs

- Changing the multiple arrows into two.



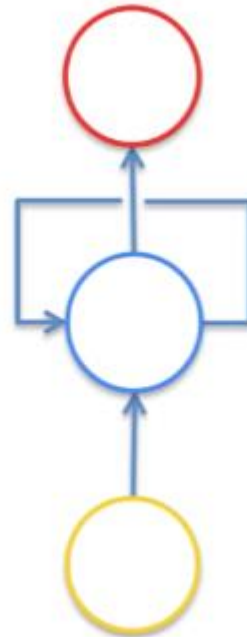
Representation of the RNNs

- ❑ Twisting it to make it vertical because that's the standard representation.



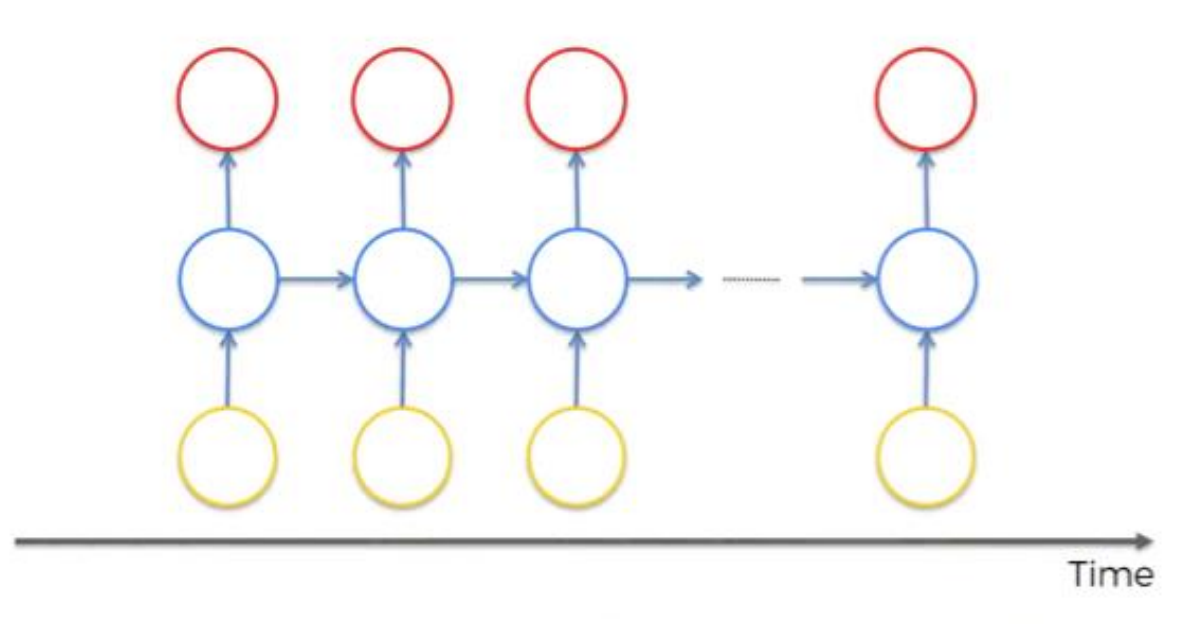
Representation of the RNNs

- Adding a line, which represents a temporal loop.
- This is an old-school representation of RNNs and basically means that this hidden layer not only gives an output but

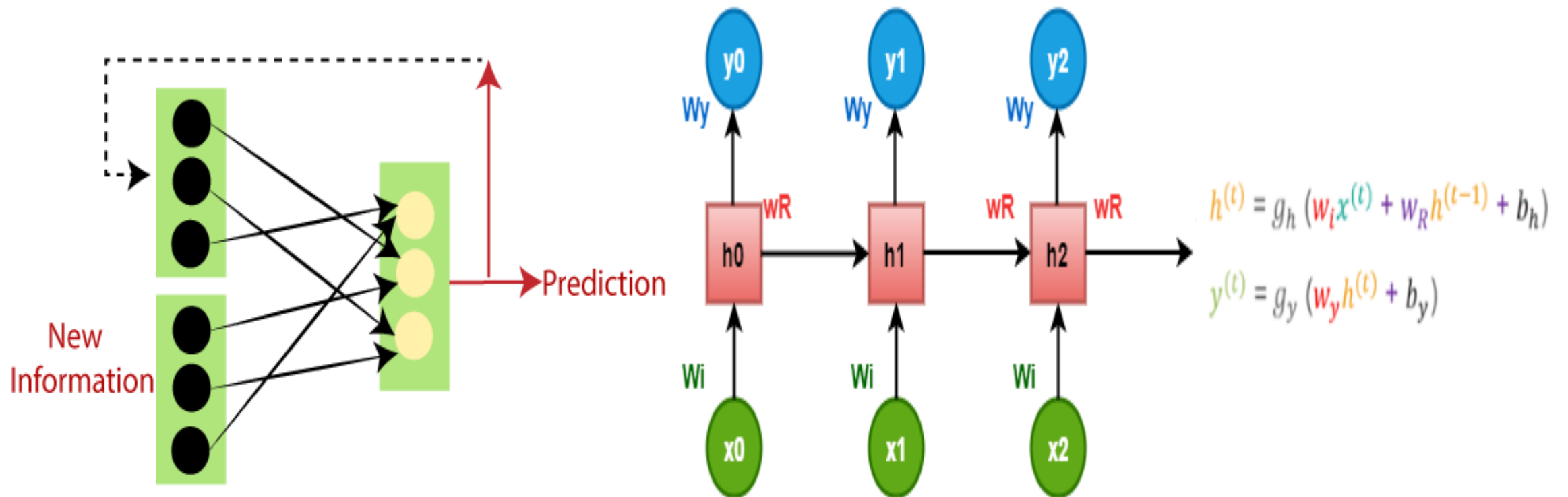


Representation of the RNNs

- Unrolling the temporal loop and representing RNNs in a new way.
- Now each circle represents not only one neuron, but a whole layer of neurons.

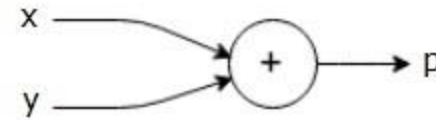


Representation of the RNNs

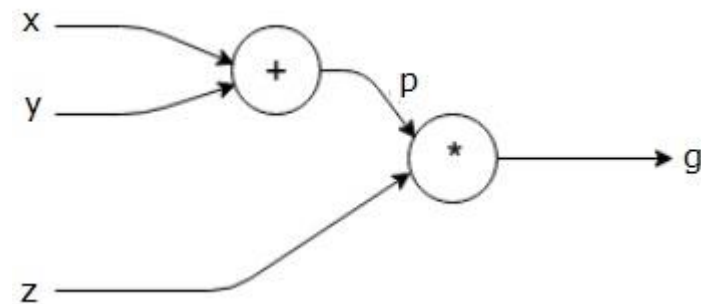


Computational Graph

- A computational graph is defined as a directed graph where the nodes correspond to mathematical operations.
- Computational graphs are a way of expressing and evaluating a mathematical expression.
- $p = x + y$

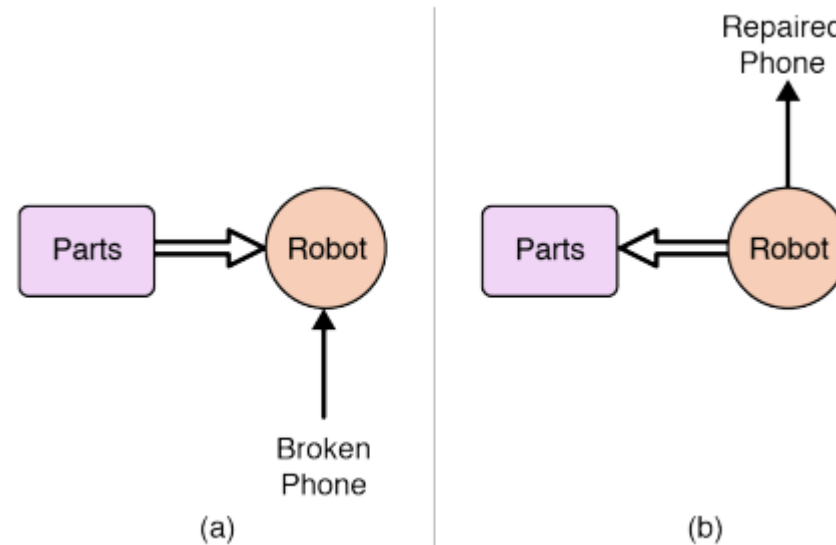


- $g = (x + y) * z$



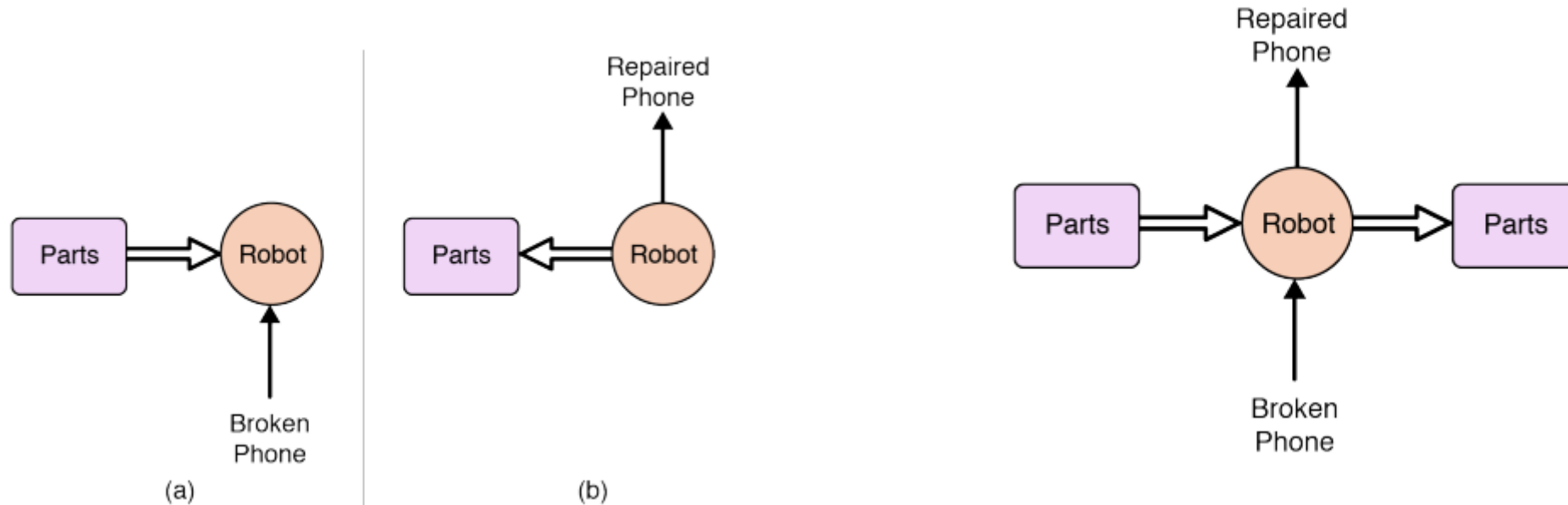
RNN

- ❑ Our robot that fixes cell phones with the help of a box of parts.
- ❑ (a) The robot is given a broken phone and a box of parts.
- ❑ (b) The robot returns to us the repaired phone, and it updates the contents of the parts box to match the parts that were removed and others that may have been added.



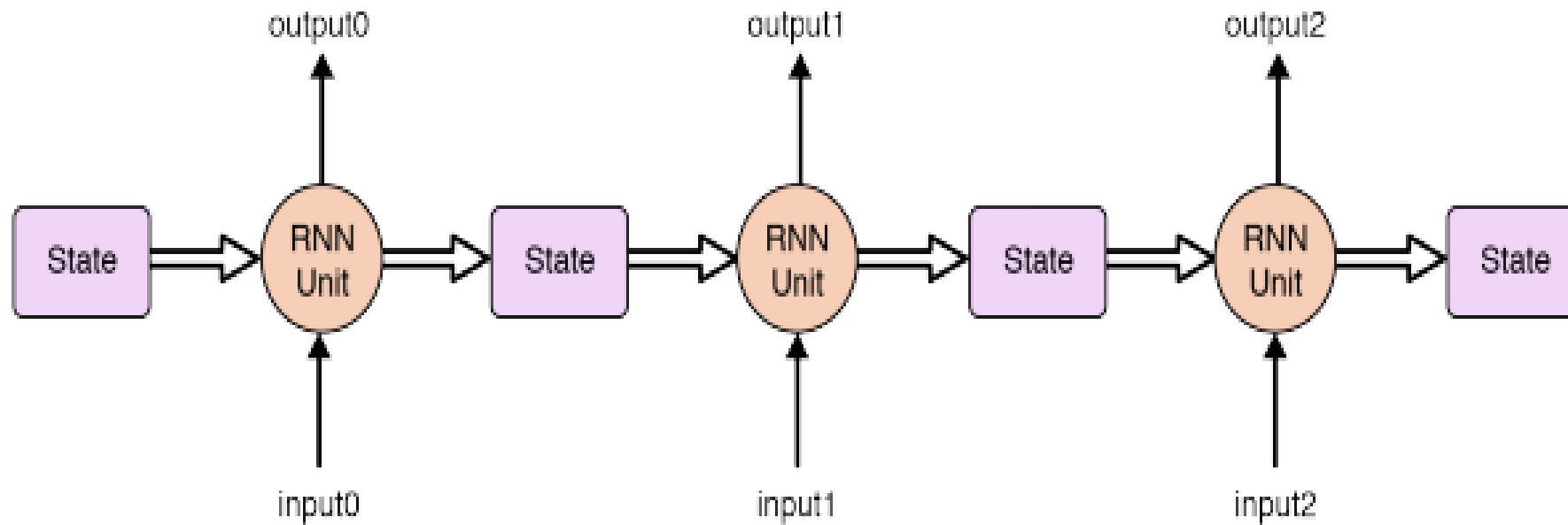
RNN

- Now let's compress the two pieces into just one diagram.
- The important things to keep in mind are that this picture represents two sequential steps, and the two boxes labeled “Parts” both refer to the same, single box.

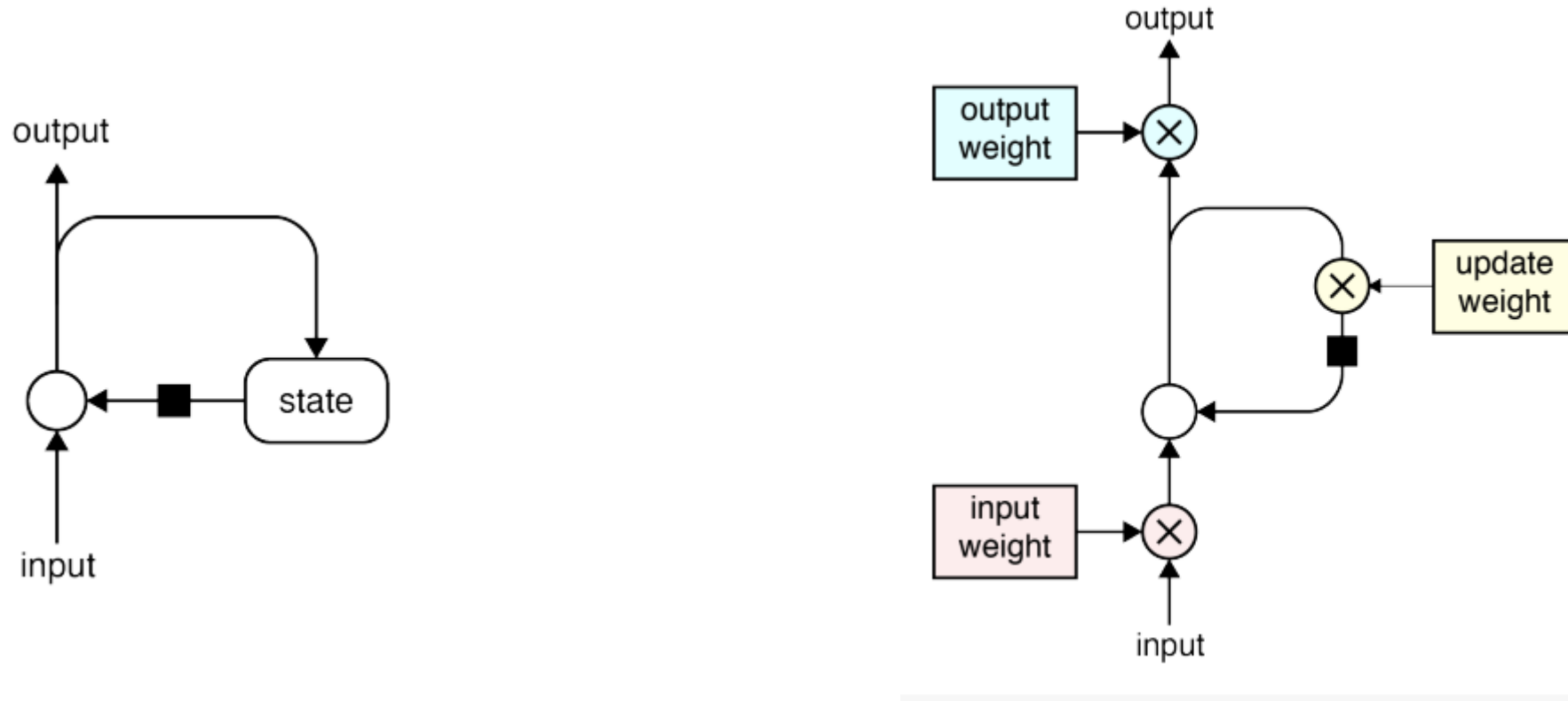


RNN

- Add more number of mobile phones

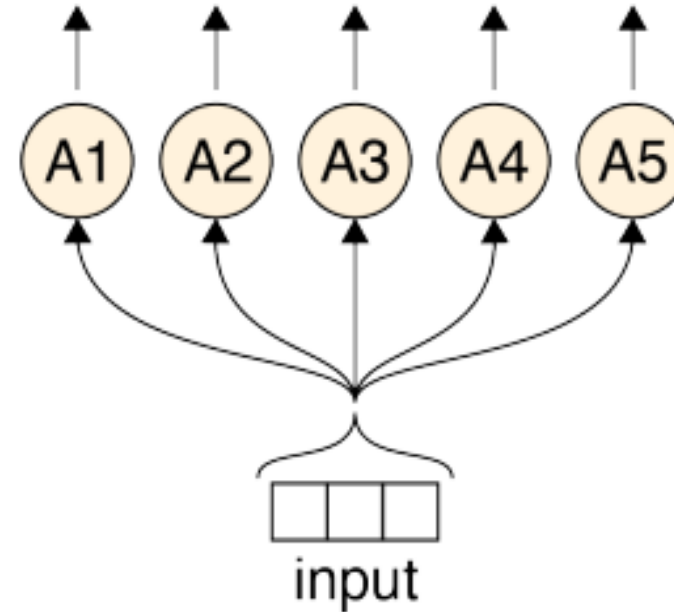
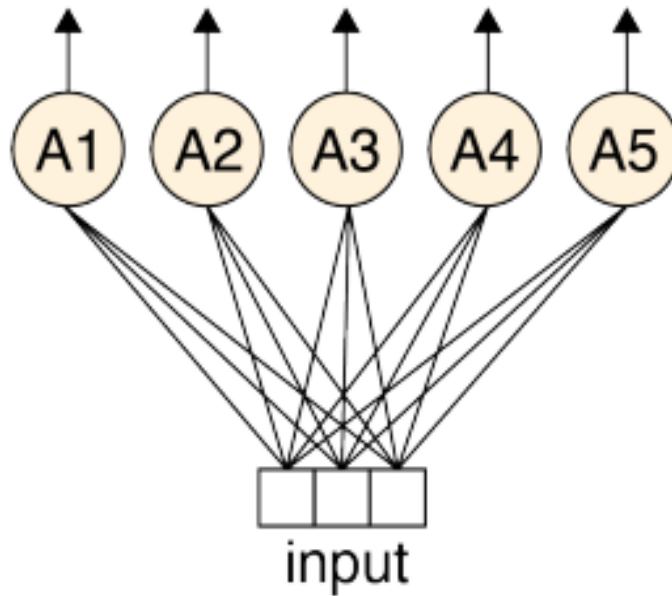


Structure of an RNN Cell

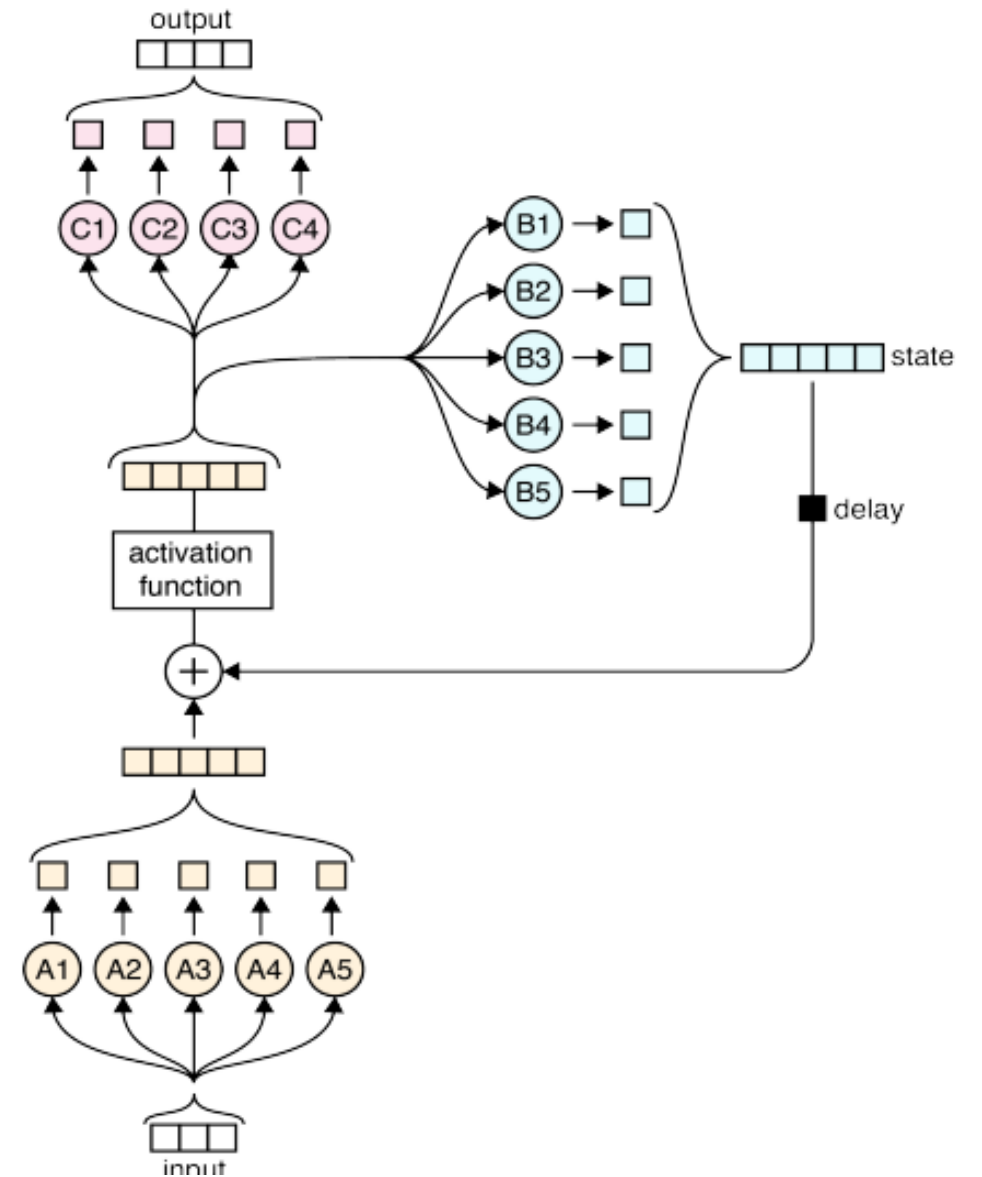


A Cell with More State

□ Traditional v/s Simplified way



A Cell with More State

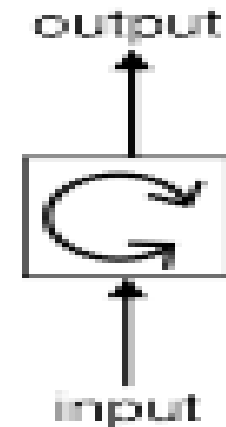
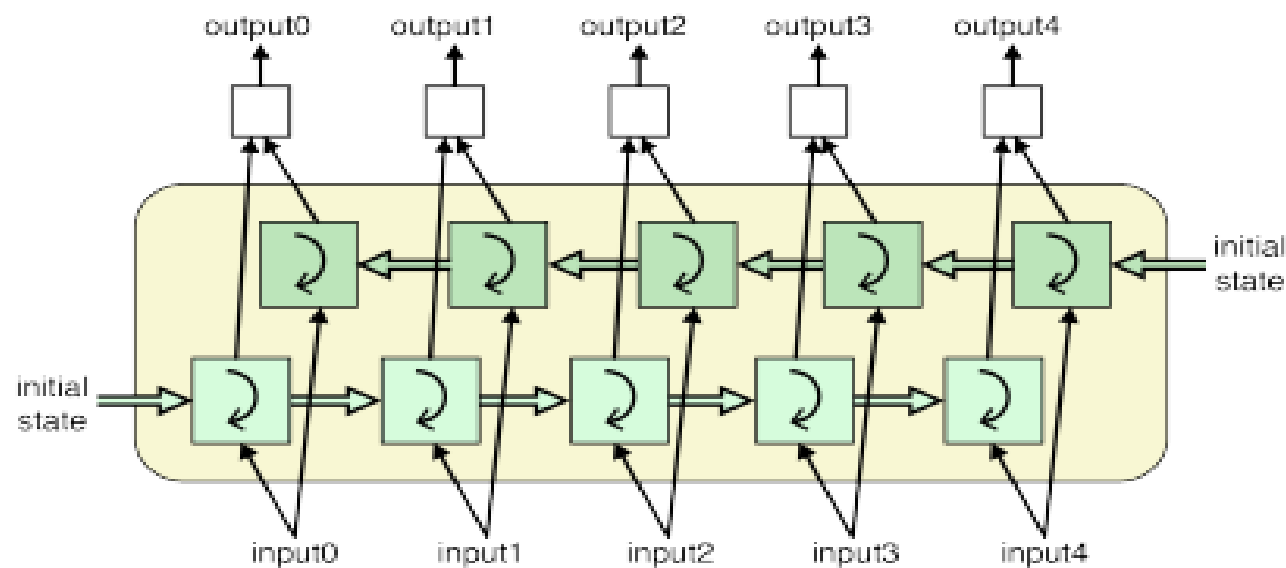


Bidirectional RNN (BRNN)

- ❑ “Charles said he needs a vacation,” figure out who is referred to by the pronoun “he.”
- ❑ “Saying, ‘I need a vacation’, Charles sat down.” who “I” refers to.
- ❑ scanning the sentence **from finish to start** would let us know it.
- ❑ This idea led to the introduction of the **Bidirectional RNN or BRNN**
- ❑ As the name implies, this network runs the input in both directions at once.
- ❑ Of course this can only work in situations where we already have data all the way to the end of the chunk we’re trying to analyze, so it’s not applicable to every situation

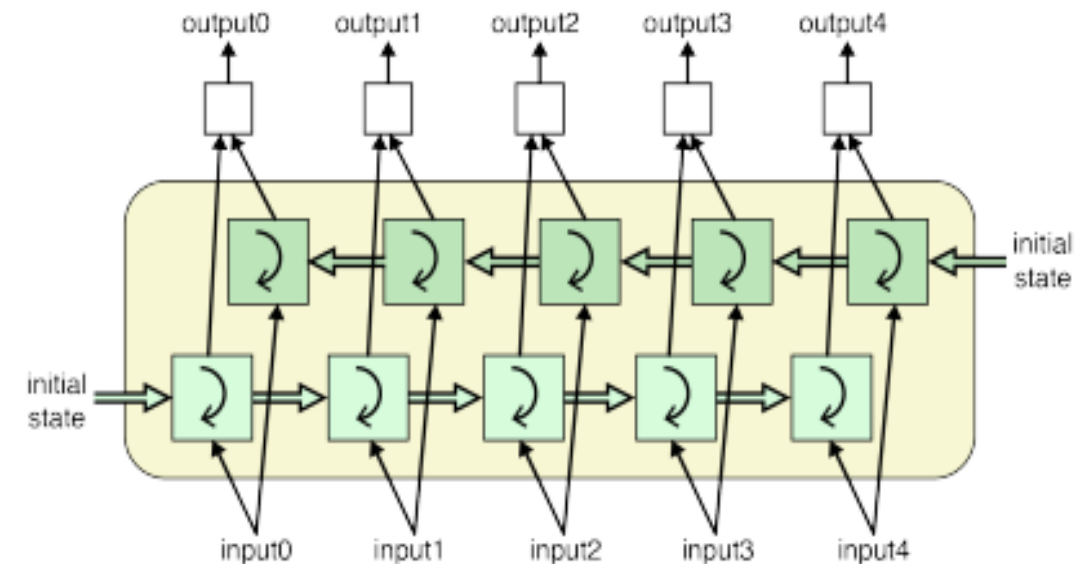
Bidirectional RNN (BRNN)

- ❑ A bidirectional RNN is two RNNs running together.
- ❑ One of them is given the time steps in the usual order, from first to last.
- ❑ The other is given the time steps in the opposite order, from last to first.



Bidirectional RNN

- ❑ To get started, the value of time step 0 is handed to the “forward” unit (light green)
- ❑ while simultaneously time step 4 is handed to the “backward” unit (dark green).
- ❑ When both LSTMs have produced their outputs, they arrive at the white boxes at the top of the figure.
- ❑ The two values at each white box are produced at different times, so the first one just sits until the second one arrives.
- ❑ Now we have two values coming into each of the square boxes at the top.
- ❑ The square boxes combine their inputs in whatever way we choose.
- ❑ Typical options are to **add** them together, **average** them, **multiply** them, or create a **2-element tensor** (that is, a list) by appending one value after the other.
- ❑ Those values can then go on to act as outputs of the network, or inputs to any other layer.



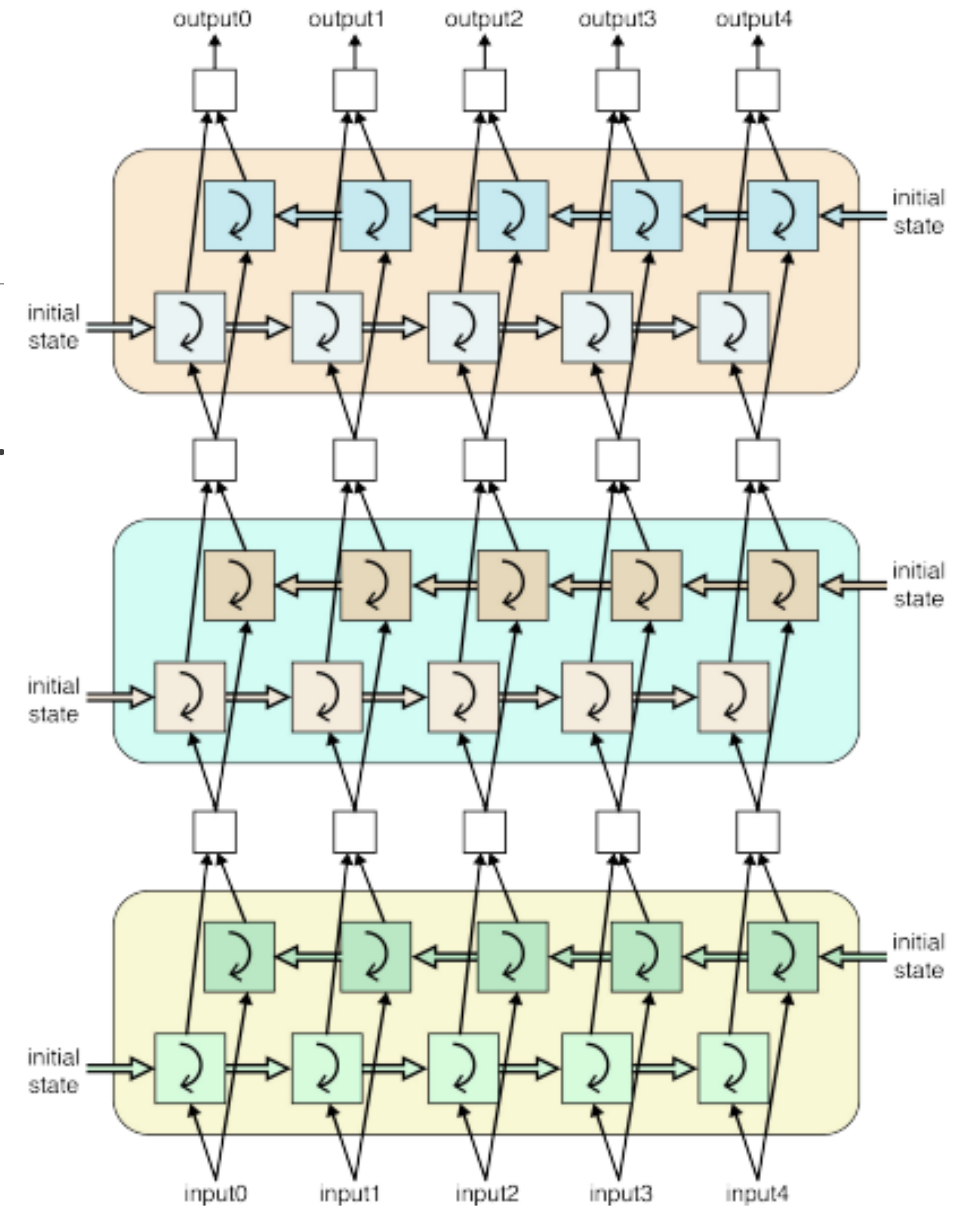
Deep BRNN

- We say this is “deep”

because there are multiple such layers, each feeding the next.

- It’s “bidirectional”

because in each layer there is both a forward and backward unit.



An Example -

- ❑ RNN using three collections of Sherlock Holmes short stories by Arthur Conan Doyle

To Sherlock Holmes she is always THE woman. I have seldom heard

To Sherlock Holmes she is always THE wom

Sherlock Holmes she is always THE woman.

rlock Holmes she is always THE woman. I

ck Holmes she is always THE woman. I hav

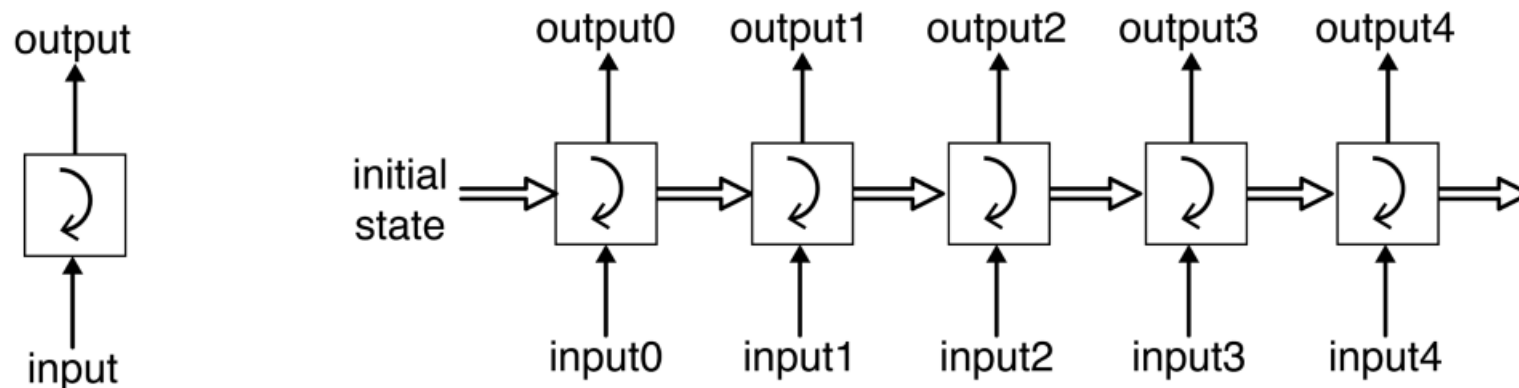
Holmes she is always THE woman. I have s

mes she is always THE woman. I have seld

- ❑ The source text (top line) is chopped up into 40-character pieces, **starting at every third character.**
- ❑ Each row below the first is a single sample of training data, presented to the RNN as 1 feature with 40 time steps.

Training an RNN

- An RNN is just a packaged-up cluster of small artificial neurons and other pieces.
- Since they learn by backprop, it stands to reason that the whole unit should learn by backprop, too.
- The usual approach is to “unroll” the RNN unit first, so we can more easily see what we’re dealing with.
- When we create our RNN we tell it how many time steps we’ll be providing (typically, every feature must have the same number of time steps, so that the tensor is a complete block of numbers)



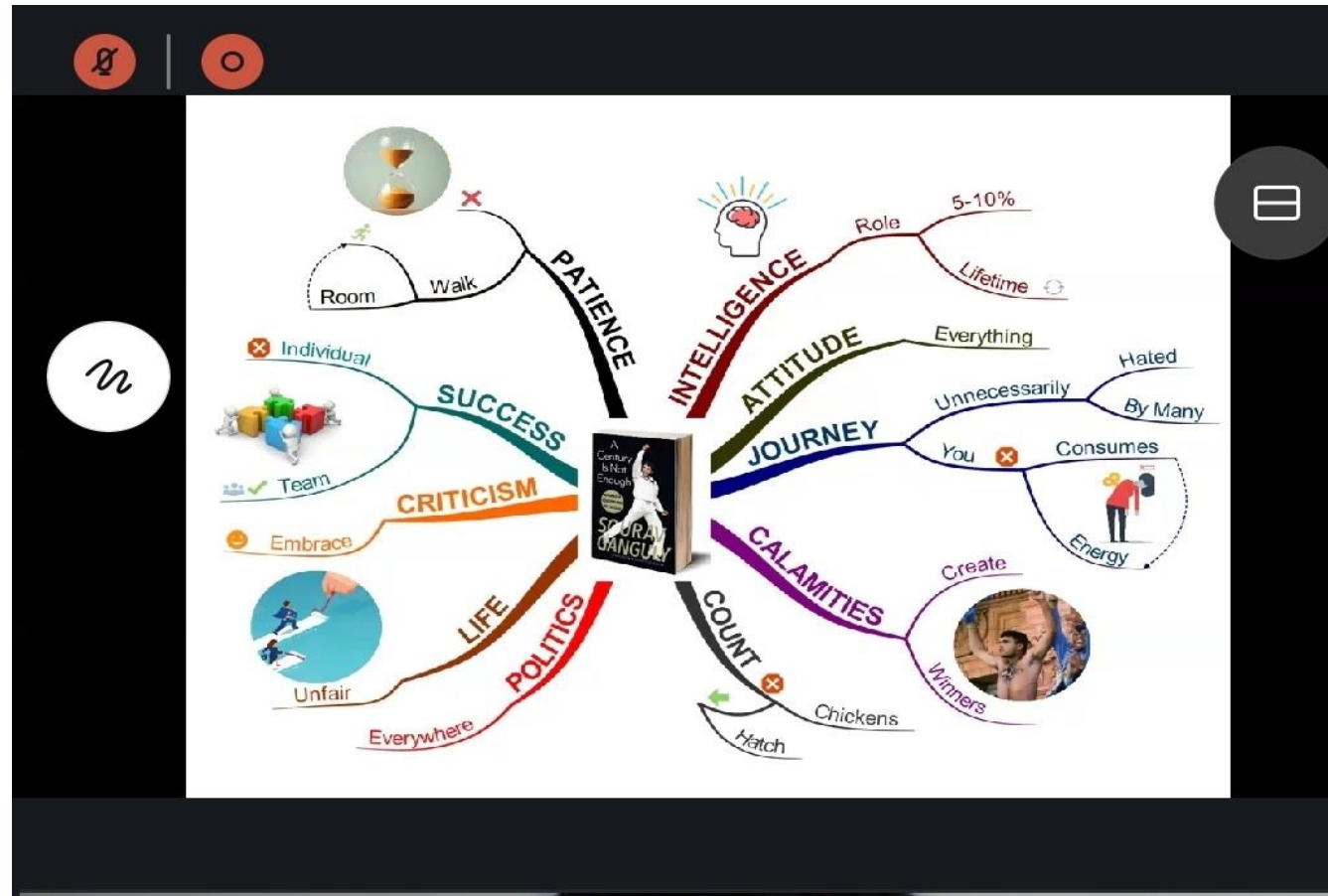
Training an RNN

- ❑ We can apply backprop to, just like any neural network, just by pushing the error gradients backwards.
- ❑ The only difference is that we need to keep in mind that every instance of the RNN represents **the same unit with the same internal weights**.
- ❑ This modified version of backprop is called **backpropagation through time, or BPTT**.

Issues with BPTT

- ❑ Vanishing gradient problem
- ❑ Exploding gradient problem





Long Term Dependency Problem

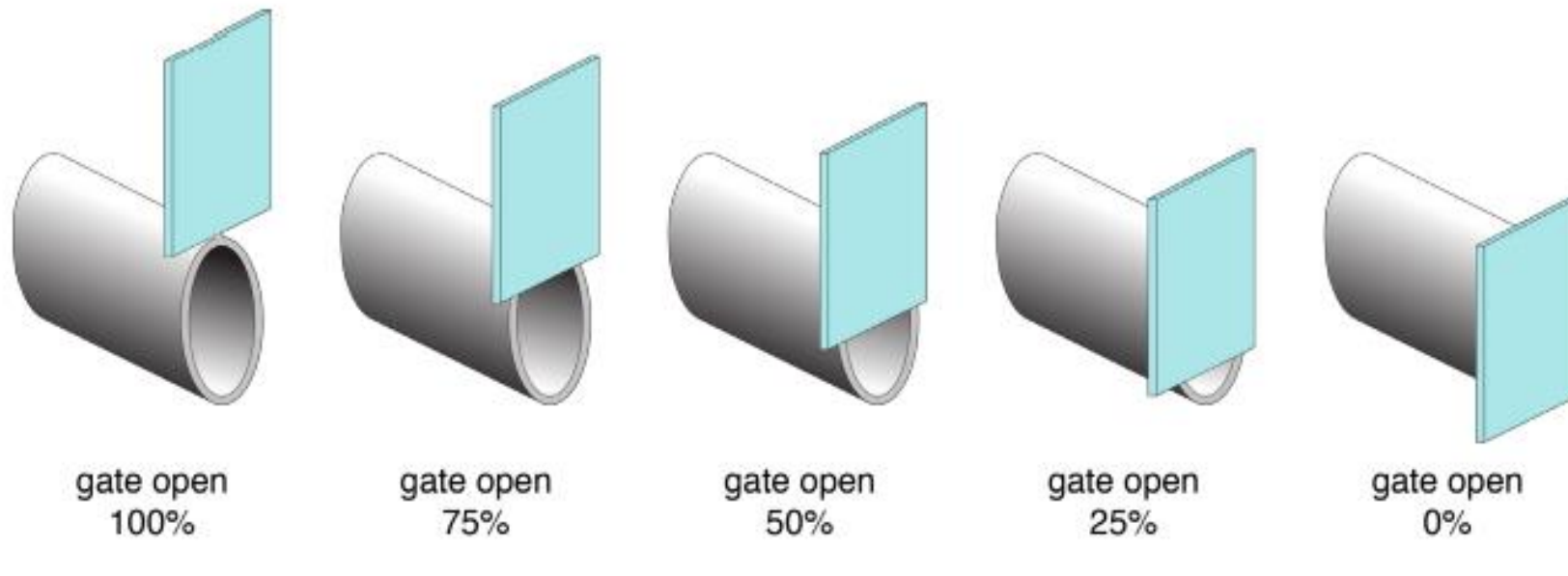
- “Bob said he was hungry,”
- “When Bob saw his neighbor’s two cats outside the garage, watching him, he ignored their focused stares and continued with his elaborate workout regimen...”
and after many words concludes with,
“...and when it was all over, they were still there, watching him.”
- We can solve both the gradient problem and the long-term dependency problem by using **RNNs with fancy internals.**

Long Short-Term Memory (LSTM)

- A cell has some persistent, or long-term memory in the form of its state.
- The cell also has some short-term memory in the form of the neuron outputs.
- The goal of the LSTM is to take some of those short-term, fleeting values and give them a longer lifespan, allowing them to contribute to future calculations.
- Thus the short-term memory is being given a longer life, resulting in the name “long short-term memory,” or LSTM.
- It might be helpful to think of this instead as “persistent short-term memory.”

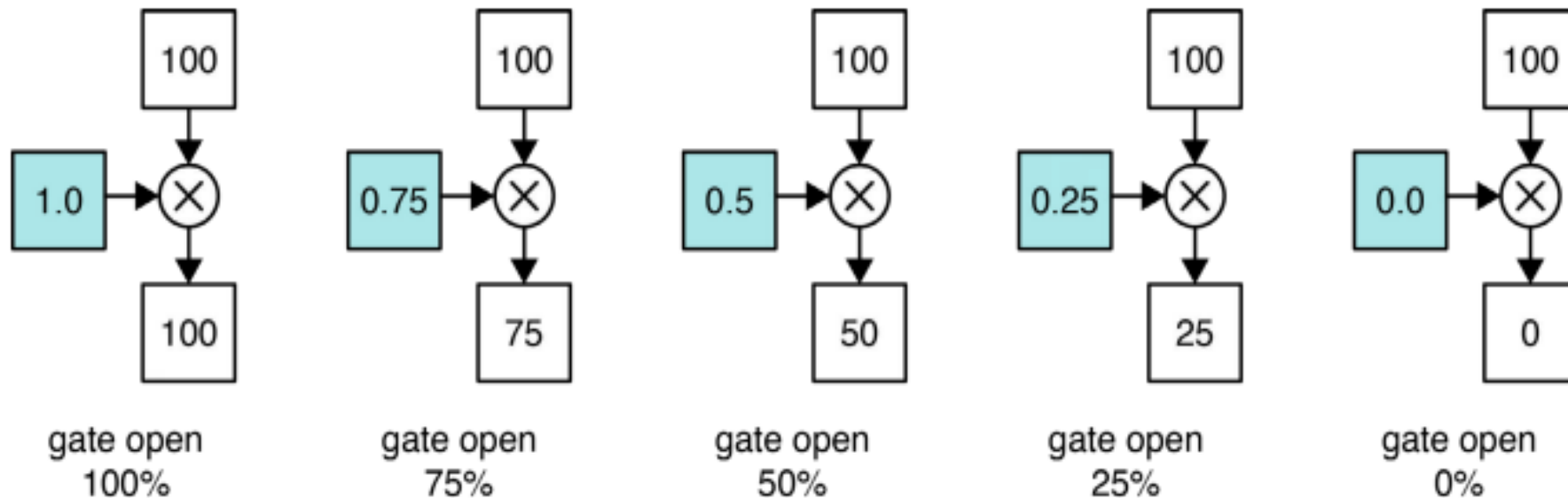
Gates

- A key idea in the LSTM is a mechanism called a gate.
- We can visualize a gate roughly as a physical gate that controls the flow of water out of a pipe.



Gates

- We can easily implement a gate in a program just by multiplying a starting value (the amount of water in the pipe) by the gate value (its position).



- When we use gates in this way, we restrict the gate values to the range 0 to 1.

Gates

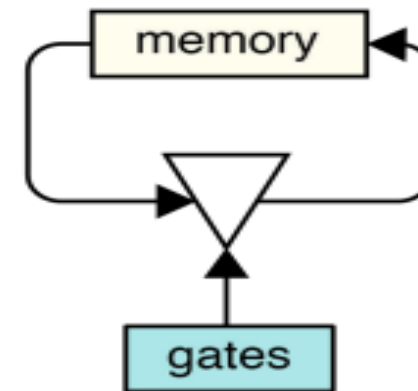
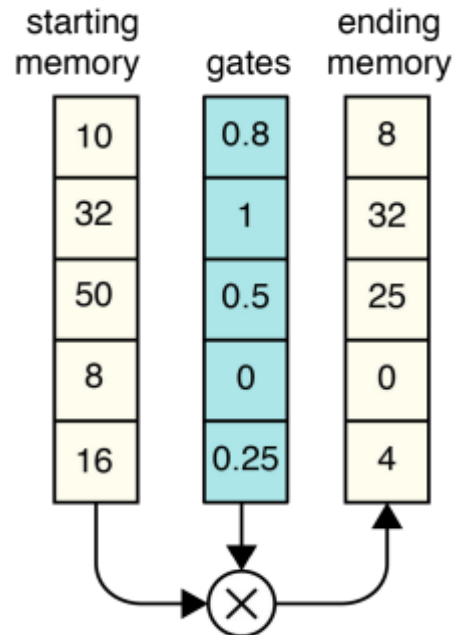
An LSTM uses gates for three purposes:

- 1) Forgetting
- 2) Remembering
- 3) Selecting

The remember and select gates are also called the input and output gates.

Forgetting

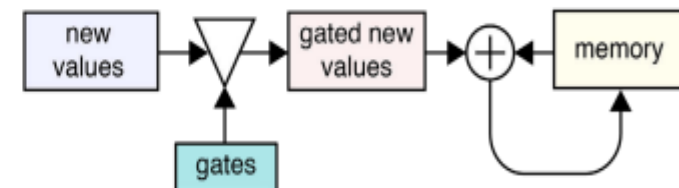
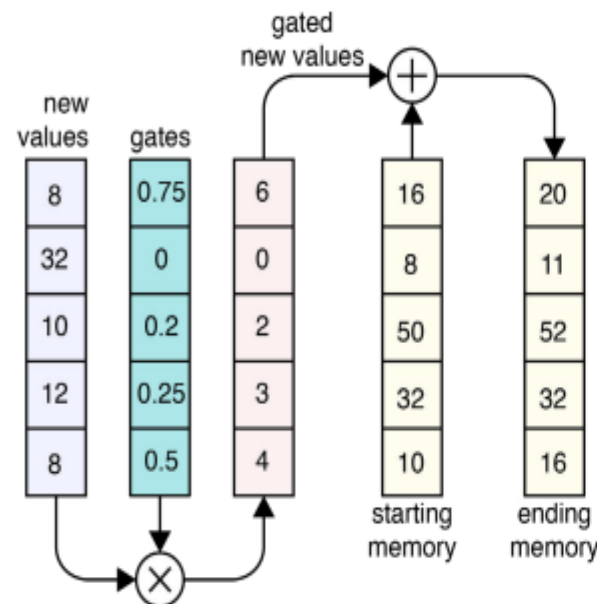
- ❑ To forget a number means that we push it towards 0.
- ❑ When it reaches 0, it's completely forgotten.
- ❑ We can selectively forget a value by multiplying it with a gate value, which we know is always between 0 and 1.



Remembering

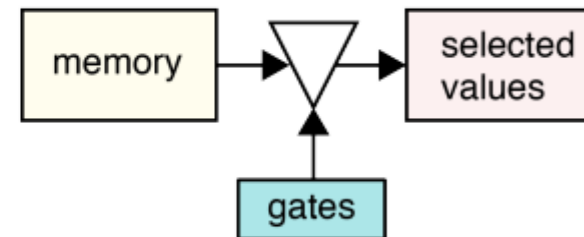
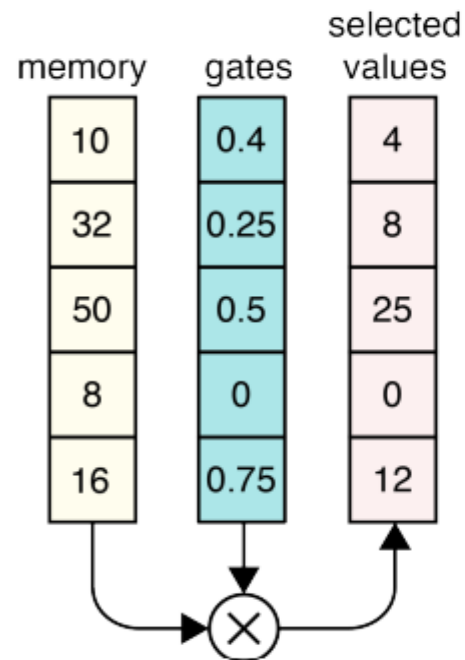
- The act of remembering involves two steps.
- First, we determine how much of each new value we want to remember.
- Then to remember the gated values.

- First, we use gates to remember
And then
- Merely add them in to the
existing contents of the memory.



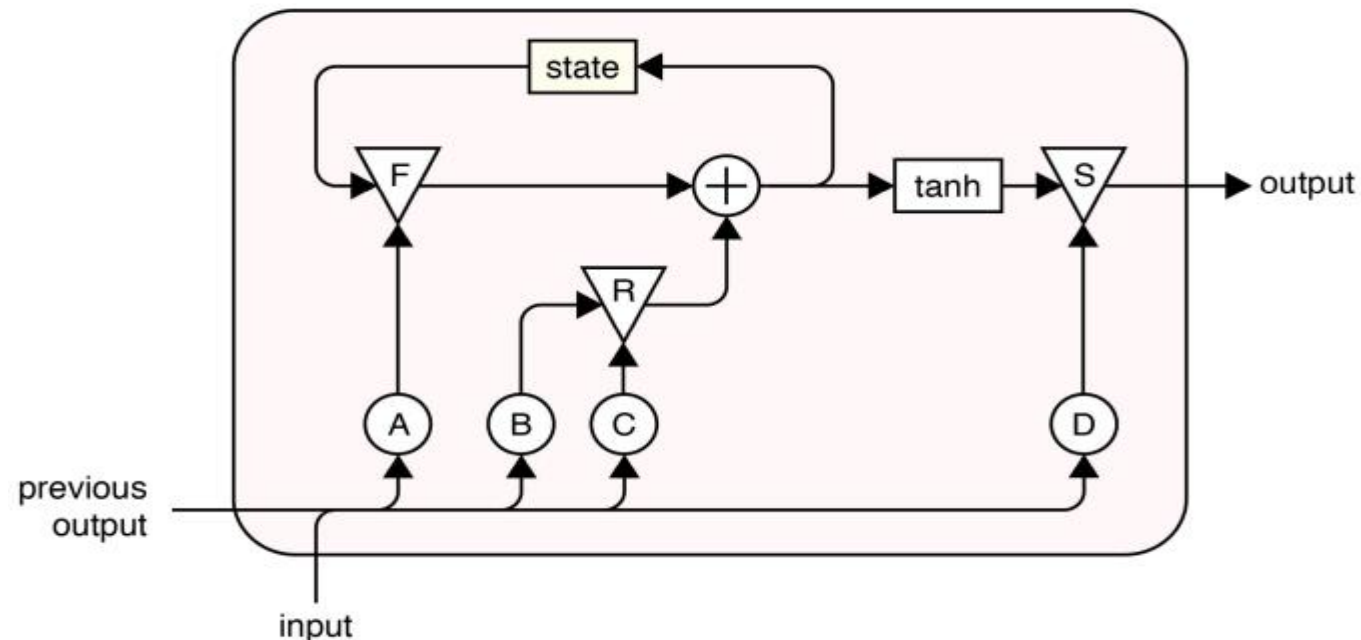
Select

- ❑ Finally, to select from memory we just determine how much of each element we want to use.
- ❑ We apply gates to the memory elements, and the results are a list of scaled memories.

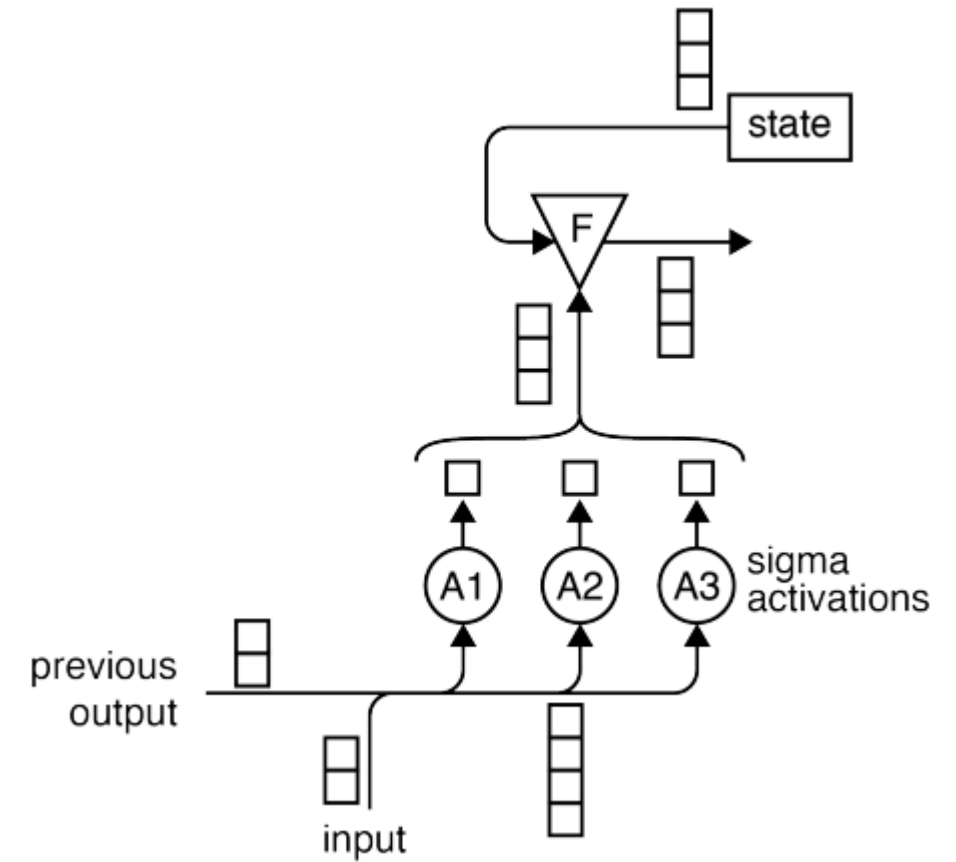
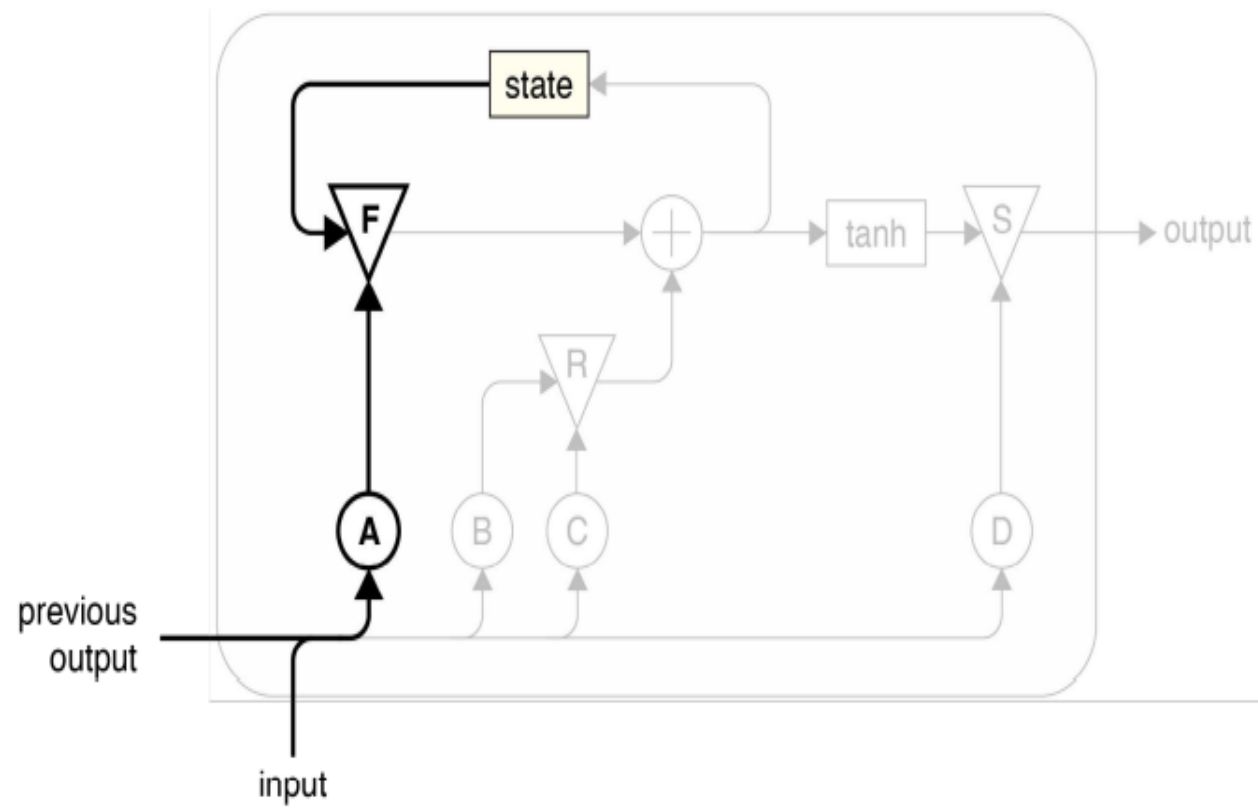


Architecture of LSTM

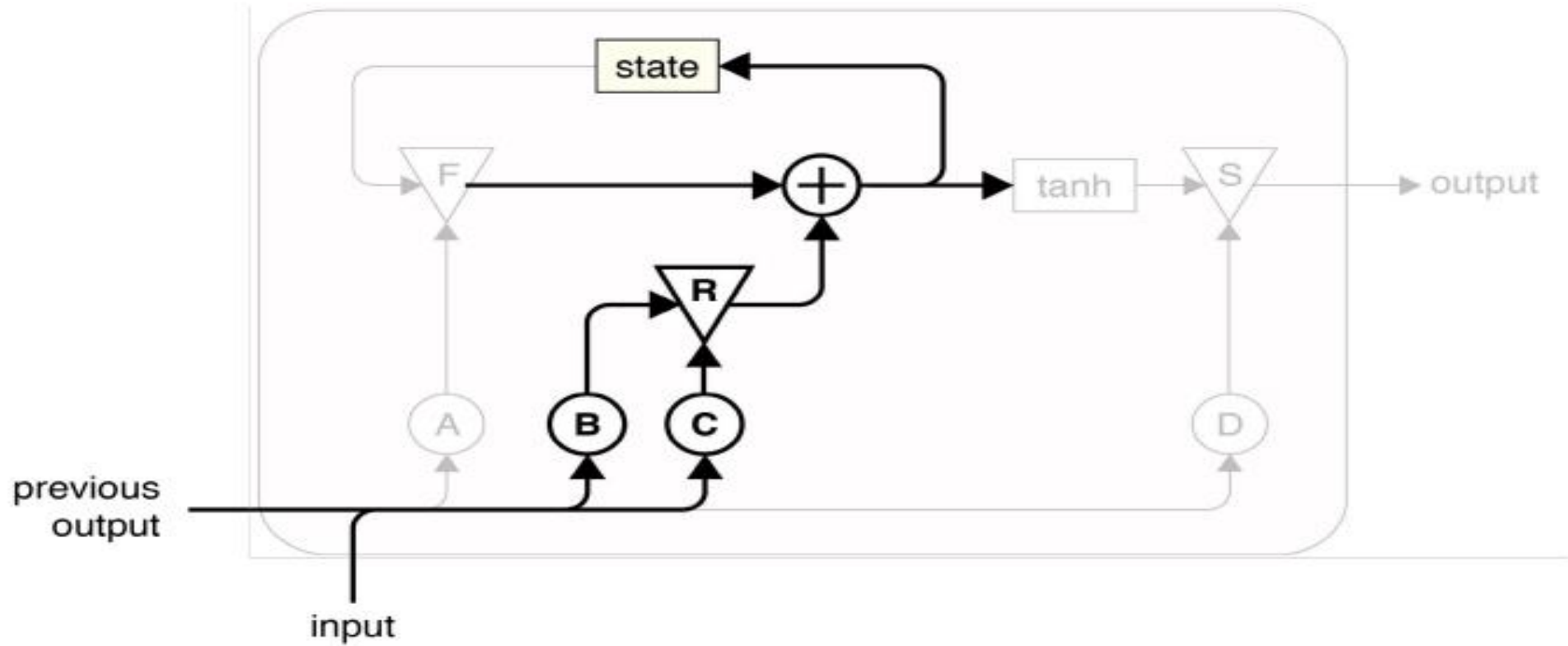
- ❑ LSTM unit uses the gating steps to manage its internal memory.
- ❑ The gating operations give the cell a lot of control over what it remembers and forgets over time, so it can manage its internal cell memory in the most effective way.



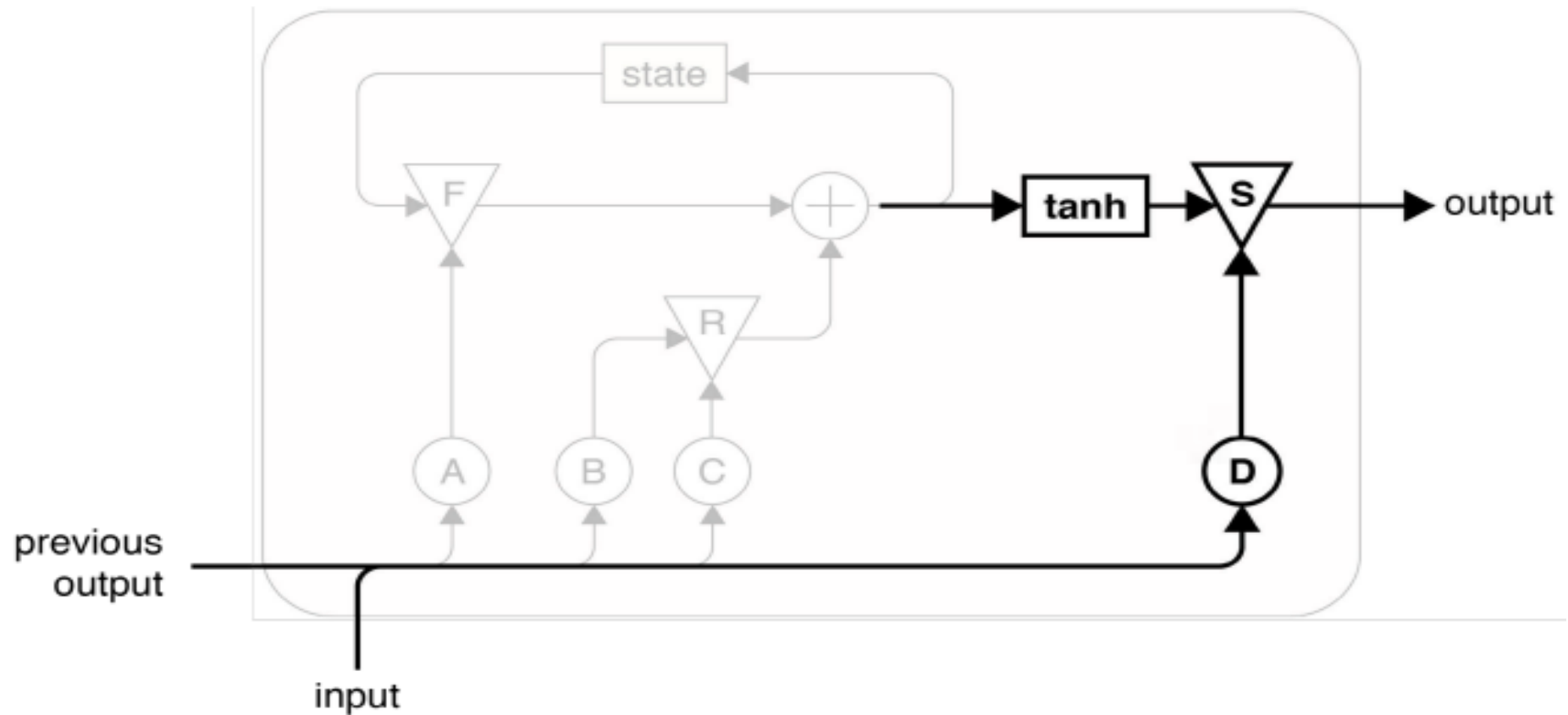
Forgetting LSTM



Remembering LSTM



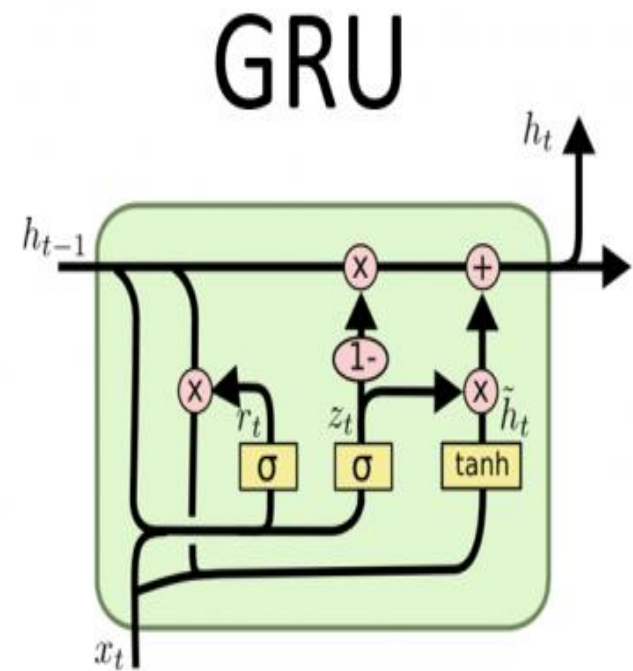
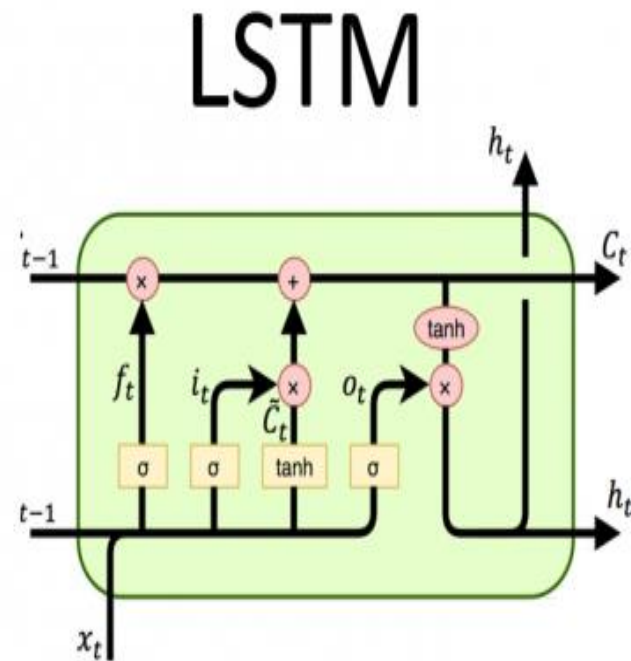
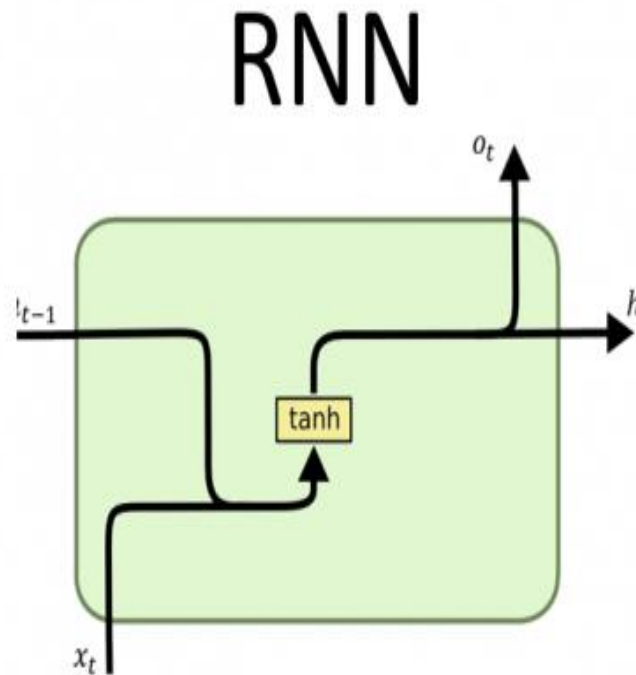
Selecting LSTM



Gated Recurrent Unit – GRU

- ❑ One of the more famous variants of the LSTM is called the Gated Recurrent Unit, or GRU
- ❑ The GRU is like an LSTM but with some simplifications.
- ❑ For instance, the forget and input gates are combined into a single gate.
- ❑ Since there's a bit less work to be done, a GRU can be a bit faster than an LSTM.
- ❑ It also usually produces results that are similar to the LSTM .
- ❑ When working with RNNs, it's often worthwhile to try both the LSTM and GRU to see if either provides more accurate results for a particular network and data set.

RNN, LSTM and GRU



RNN Structures

